

# PYCIP 500mg Film coated tablet

## COMPOSITION

**Active Ingredient:** Each film – coated tablet contains ciprofloxacin hydrochloride equivalent to 500 mg of ciprofloxacin.

**Other Ingredients:** Pregelatinized starch, Maize starch, Magnesium stearate, Sodium starch glycolate, Opadry white, Titanium dioxide.

## PHARMACODYNAMIC

Ciprofloxacin has *in vitro* activity against a wide range of gram-negative and gram-positive microorganisms. The bactericidal action of ciprofloxacin results from inhibition of the enzymes topoisomerase II (DNA gyrase) and topoisomerase IV, which are required for bacterial DNA replication, transcription, repair, and recombination. The mechanism of action of fluoroquinolones, including ciprofloxacin, is different from that of penicillins, cephalosporins, aminoglycosides, macrolides, and tetracyclines; therefore, microorganisms resistant to these classes of drugs may be susceptible to ciprofloxacin and other quinolones. There is no known cross-resistance between ciprofloxacin and other classes of antimicrobials. *In vitro* resistance to ciprofloxacin develops slowly by multiple step mutations. Ciprofloxacin has been shown to be active against most strains of the following microorganisms, both *in vitro* and in clinical infections as described in the indications and usage section of the package insert for PYCIP:

Aerobic gram-positive microorganisms

*Enterococcus faecalis* (Many strains are only moderately susceptible), *Staphylococcus aureus* (methicillin-susceptible strains only), *Staphylococcus epidermidis* (methicillin-susceptible strains only), *Staphylococcus saprophyticus*, *Streptococcus pneumoniae* (penicillin-susceptible strains only), *Streptococcus pyogenes*.

Aerobic gram-negative microorganisms

*Campylobacter jejuni*, *Proteus mirabilis*, *Citrobacter diversus*, *Proteus vulgaris*, *Citrobacter freundii*, *Providencia rettgeri*, *Enterobacter cloacae*, *Providencia stuartii*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Haemophilus influenzae*, *Salmonella typhi*, *Haemophilus parainfluenzae*, *Serratia marcescens*, *Klebsiella pneumoniae*, *Shigella boydii*, *Moraxella catarrhalis*, *Shigella dysenteriae*, *Morganella morganii*, *Shigella flexneri*, *Neisseria gonorrhoeae*, *Shigella sonnei*.

## PHARMACOKINETIC

- Ciprofloxacin given as an oral tablet is rapidly and well absorbed from the gastrointestinal tract after oral administration. The absolute bioavailability is approximately 70% with no substantial loss by first pass metabolism.
- The binding of ciprofloxacin to serum proteins is 20 to 40%. After oral administration, ciprofloxacin is widely distributed throughout the body. Tissue concentrations often exceed serum concentrations in both men and women, particularly in genital tissue including the prostate. The drug diffuses into the cerebrospinal fluid (CSF); however, CSF concentrations are generally less than 10% of peak serum concentrations. Low levels of the drug have been detected in the aqueous and vitreous humors of the eye.
- Four metabolites have been identified in human urine which together account for approximately 15% of an oral dose. The metabolites have antimicrobial activity, but are less active than unchanged ciprofloxacin. Ciprofloxacin is an inhibitor of human cytochrome P450 1A2 (CYP1A2) mediated metabolism.
- The serum elimination half-life in subjects with normal renal function is approximately 4 hours. Approximately 40 to 50% of an orally administered dose is excreted in the urine as unchanged drug. The urinary excretion of ciprofloxacin is virtually complete within 24 hours after dosing. Although bile concentrations of ciprofloxacin are several fold higher than serum concentrations after oral dosing, only a small amount of the dose administered is recovered from the bile as unchanged drug. Approximately 20 to 35% of an oral dose is recovered from the feces within 5 days after dosing. This may arise from either biliary clearance or transintestinal elimination.

## INDICATIONS

Consideration should be given to official guidance on the appropriate use of antibacterial agents.

Ciprofloxacin are indicated for the treatment of the following infections:

- Infections of the respiratory tract, middle ear, paranasal sinuses, eyes, kidneys and/or efferent urinary tract, biliary tract, gastrointestinal, skin and soft tissues, bones and joints due to sensitive organisms.
- Peritonitis.
- Reduction of disease following exposure to aerosolised *Bacillus anthracis*.
- For children 5 - 17 years, broncho-pulmonary infections of cystic fibrosis associated with *Pseudomonas aeruginosa*.
- For children 1 - 17 years, 2nd-line and 3rd-line treatment of complicated UTI and pyelonephritis due to *E. coli*.

**The following indications, should be restricted:**

- Acute bacterial rhinosinusitis\*
- Acute exacerbation of chronic obstructive pulmonary disease including chronic

bronchitis\*

- Nosocomial pneumonia/ Hospital-acquired pneumonia\*
- Acute otitis media\*
- External otitis\*
- Endocarditis\*
- Infection of cerebrospinal fluid\*
- Meningitis\*
- Septicaemia\*
- Uncomplicated acute cystitis/ uncomplicated cystitis\*
- Prevention of exacerbations in women with recurring urinary tract infection\*
- Prevention of infection in surgical procedures in the urogenital system\*#.
- Pre-operative preparations for chronic cholesteatomatous otitis and chronic otitis spreading to bone\*

**\*Ciprofloxacin should be only used:**

- + When *Pseudomonas* is considered and the patient is allergic to antipseudomonal penicillins/ cephalosporins.
- + For resistant organisms with no other alternative antibiotics available.

# Ciprofloxacin should not be used > 24 hours post operation.

## CONTRA-INDICATIONS

Ciprofloxacin tablets are contraindicated in persons with a history of hypersensitivity to ciprofloxacin, any member of the quinolone class of antimicrobial agents, or any of the product components.

## SPECIAL WARNINGS AND PRECAUTIONS FOR USE

### Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population.

Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis).

In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Side Effects). Treatment of these patients with ciprofloxacin should be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

### Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (Musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ciprofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

### Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb (so should be appropriately treated (e.g. immobilisation)). Corticosteroids should not be used if signs of tendinopathy occur.

### Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Side Effects).

### Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been

associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

#### **Psychiatric reactions**

Psychiatric reactions may occur even after the first administration of fluoroquinolones, including PYCIP. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Side Effect'). In the event that the patient develops these reactions, PYCIP should be discontinued and appropriate measures instituted. Caution is recommended if PYCIP is to be used in psychotic patients or in patients with a history of psychiatric disease.

#### **PRECAUTIONS**

##### **General**

Crystals of ciprofloxacin have been observed rarely in the urine of human subjects but more frequently in the urine of laboratory animals, which is usually alkaline. Crystalluria related to ciprofloxacin has been reported only rarely in humans because human urine is usually acidic. Alkalinity of the urine should be avoided in patients receiving ciprofloxacin. Patients should be well hydrated to prevent the formation of highly concentrated urine.

##### **Central Nervous System**

Quinolones, including ciprofloxacin, may also cause central nervous system (CNS) events, including: nervousness, agitation, insomnia, anxiety, nightmares or paranoia.

#### **INTERACTION WITH OTHER DRUGS**

- As with some other quinolones, concurrent administration of ciprofloxacin with theophylline may lead to elevated serum concentrations of theophylline and prolongation of its elimination half-life. This may result in increased risk of theophylline-related adverse reactions. If concomitant use cannot be avoided, serum levels of theophylline should be monitored and dosage adjustments made as appropriate.
  - Some quinolones, including ciprofloxacin, have also been shown to interfere with the metabolism of caffeine. This may lead to reduced clearance of caffeine and a prolongation of its serum half-life.
  - Concurrent administration of a quinolone, including ciprofloxacin, with multivalent cation-containing products such as magnesium/aluminum antacids, sucralate, didanosine chewable/buffered tablets or pediatric powder, other highly buffered drugs, or products containing calcium, iron, or zinc may substantially decrease its absorption, resulting in serum and urine levels considerably lower than desired.
  - Histamine H<sub>2</sub>-receptor antagonists appear to have no significant effect on the bioavailability of ciprofloxacin.
  - Altered serum levels of phenytoin (increased and decreased) have been reported in patients receiving concomitant ciprofloxacin.
  - The concomitant administration of ciprofloxacin with the sulfonylurea glyburide has, on rare occasions, resulted in severe hypoglycemia.
  - Some quinolones, including ciprofloxacin, have been associated with transient elevations in serum creatinine in patients receiving cyclosporine concomitantly.
  - Quinolones, including ciprofloxacin, have been reported to enhance the effects of the oral anticoagulant warfarin or its derivatives. When these products are administered concomitantly, prothrombin time or other suitable coagulation tests should be closely monitored.
  - Probenecid interferes with renal tubular secretion of ciprofloxacin and produces an increase in the level of ciprofloxacin in the serum. This should be considered if patients are receiving both drugs concomitantly.
  - Renal tubular transport of methotrexate may be inhibited by concomitant administration of ciprofloxacin potentially leading to increased plasma levels of methotrexate. This might increase the risk of methotrexate associated toxic reactions. Therefore, patients under methotrexate therapy should be carefully monitored when concomitant ciprofloxacin therapy is indicated.
  - Metoclopramide significantly accelerates the absorption of oral ciprofloxacin resulting in shorter time to reach maximum plasma concentrations. No significant effect was observed on the bioavailability of ciprofloxacin.
  - Non-steroidal anti-inflammatory drugs (but not acetyl salicylic acid) in combination of very high doses of quinolones have been shown to provoke convulsions in pre-clinical studies.
- DOSAGE AND ADMINISTRATION**
- Respiratory tract infection** 250-500 mg bd.
- Acute and uncomplicated urinary tract infection** 125-250 mg bd.
- Cystitis in women** 250 mg single dose.
- Complicated urinary tract infection** 250-500 mg bd.
- Extragenital gonorrhea** 125 mg bd.
- Acute and uncomplicated gonorrhea** 250 mg single dose.
- Diarrhea** 500 mg bd.
- Streptococcal pneumonia, recurrent infections in cystic fibrosis, bone and joint infections, septicemia, peritonitis** 750 mg bd.
- Inhalational anthrax (post-exposure)** 500 mg bd.

**Other infections** 500 mg bd. ROUTE

OF ADMINISTRATION: Oral.

#### **PREGNANCY AND LACTATION**

##### **Pregnancy**

The data that are available on administration of ciprofloxacin to pregnant women indicates no malformative or feto/neonatal toxicity of ciprofloxacin. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity

As a precautionary measure, it is preferable to avoid the use of ciprofloxacin during pregnancy.

##### **Lactation**

Ciprofloxacin is excreted in breast milk. Due to the potential risk of articular damage, ciprofloxacin should not be used during breast-feeding.

##### **EFFECTS ON ABILITY TO DRIVE AND USE MACHINES**

Due to its neurological effects, ciprofloxacin may affect reaction time. Thus the ability to drive or to operate machinery may be impaired.

##### **SIDE EFFECT**

**GENERAL DISORDERS AND ADMINISTRATIVE SITE CONDITIONS\*:** headache, abdominal pain/discomfort, foot pain, pain, pain in extremities.

**CARDIOVASCULAR:** palpitation, atrial flutter, ventricular ectopy, syncope, hypertension, angina pectoris, myocardial infarction, cardiopulmonary arrest, cerebral thrombosis, phlebitis, tachycardia, migraine, hypotension.

**NERVOUS SYSTEM DISORDERS\*:** restlessness, dizziness, lightheadedness, insomnia, irritability, tremor, ataxia, convulsive seizures, lethargy, drowsiness, weakness, malaise, anorexia, paresthesia, abnormal gait, grand mal convulsion, bad taste.

**PSYCHIATRIC DISORDERS\*:** nightmares, hallucinations, manic reaction, phobia, depersonalization, depression.

**Rare:** Depression (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

**Very Rare:** Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide)

**GASTROINTESTINAL:** painful oral mucosa, oral candidiasis, dysphagia, intestinal perforation, gastrointestinal bleeding, cholestatic jaundice, hepatitis.

**HEMIC/LYMPHATIC:** lymphadenopathy, petechia.

**METABOLIC/NUTRITIONAL:** amylase increase, lipase increase

**MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS\*:** arthralgia or back pain, joint stiffness, achiness, neck or chest pain, flare up of gout.

**RENAL/UROGENITAL:** interstitial nephritis, nephritis, renal failure, polyuria, urinary retention, urethral bleeding, vaginitis, acidosis, breast pain.

**RESPIRATORY:** dyspnea, epistaxis, laryngeal or pulmonary edema, hiccup, hemoptysis, bronchospasm, pulmonary embolism.

**SKIN/HYPERSENSITIVITY:** allergic reaction, pruritus, urticaria, flushing, fever, chills, angioedema, edema of the face, neck, lips, conjunctivae or hands, cutaneous candidiasis, hyperpigmentation, erythema nodosum, sweating.

**EYE DISORDERS\* & EAR AND LABYRINTH DISORDERS\*:** blurred vision, disturbed vision (change in color perception, overbrightness of lights), decreased visual acuity, diplopia, eye pain, tinnitus, hearing loss, chromatopsia.

In several instances nausea, vomiting, tremor, irritability, or palpitation were judged by investigators to be related to elevated serum levels of theophylline possibly as a result of drug interaction with ciprofloxacin.

\* Very rare cases if prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Special warnings and precautions for use).

##### **Post Marketing Experience**

Exacerbation of myasthenia gravis.

##### **OVERDOSE AND TREATMENT**

In the event of acute overdosage, reversible renal toxicity has been reported in some cases. The stomach should be emptied by inducing vomiting or by gastric lavage. The patient should be carefully observed and given supportive treatment, including monitoring of renal function and administration of magnesium, aluminum, or calcium containing antacids which can reduce the absorption of ciprofloxacin. Adequate hydration must be maintained. Only a small amount of ciprofloxacin (< 10%) is removed from the body after hemodialysis or peritoneal dialysis.

##### **SHELF LIFE AND STORAGE INSTRUCTIONS**

The expiry date of this medicinal product is printed on the box and blister foil. Do not use this medicinal product after the expiry date.

Storage condition: In a dry, cool place. Protect from light. Please store below 30°C.

**KEEP DRUG OUT OF REACH OF CHILDREN.**

**CAREFULLY READ THE PACKAGE INSERT BEFORE USE.**

**PREPARATIONS AND PACK SIZE**

PYCIP 500mg film – coated tablet appears as white film – coated caplet, with a fracture line on the side. Blister packed in PVC/aluminium blister and packed in card box (packs of 10 x 10's).

**DATE OF REVISION:** 18 April 2024.

**Manufacturer**  
**PYMEPHARCO JOINT STOCK**  
**COMPANY**  
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